

Lab 2 Implement an IAM Role

In this challenge, you will create an Identity and Access Management (IAM) role that you will use to access an Amazon Simple Storage Service (Amazon S3) bucket from an Elastic Compute Cloud (EC2) instance. First, you will create an Amazon S3 bucket to store objects, and then you will create a role to provide access to the bucket from an EC2 instance. Next, you will create an IAM policy that provides full control of the bucket, and then you will assign the policy to the role. Finally, you will create a new EC2 instance that will use the role, and then you will sign in to the EC2 instance and test the role. Note: Once you begin a challenge you will not be able to pause, save, or return to your progress. Please ensure you have set aside enough time to complete the challenge before you start.

Task1: Create an Amazon S3 Bucket

S3 -> create bucket

Create S3 bucket | S3 | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/s3/bucket/create?region=us-east-2&bucketType=general

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod51665701 @ 3167-3116-5550

Amazon S3 > Buckets > Create bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (Ohio) us-east-2

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

bucket-51665701

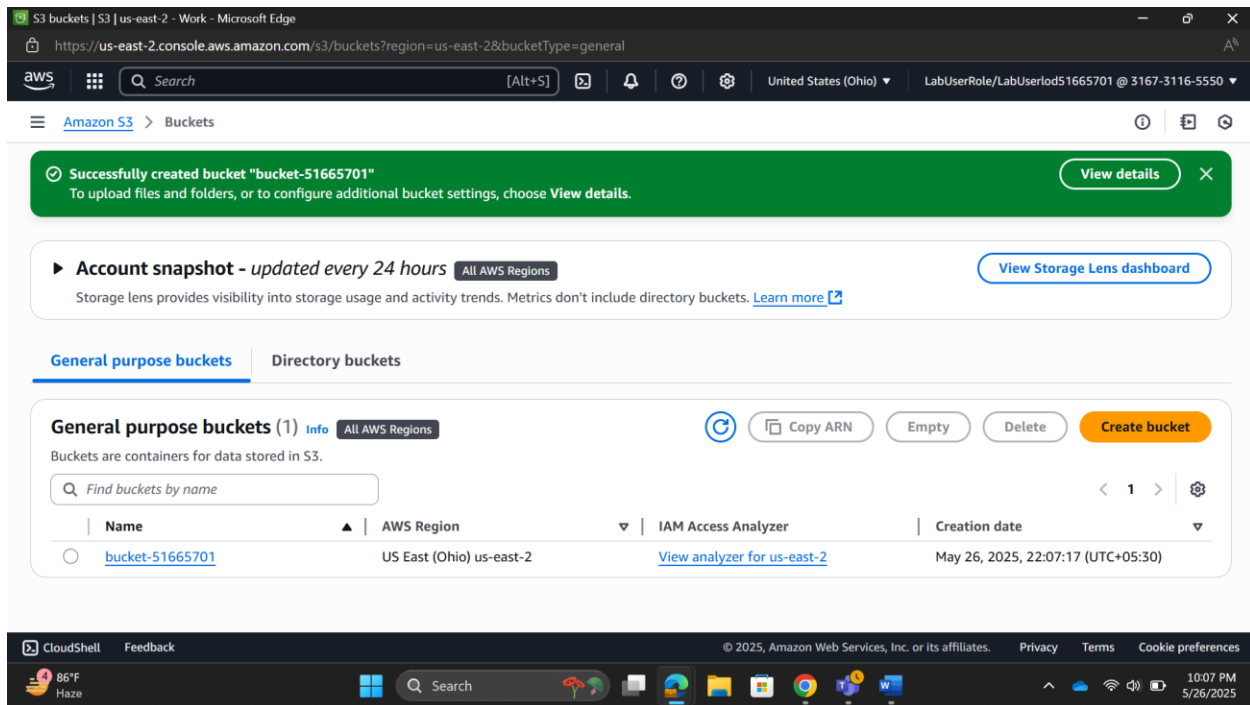
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

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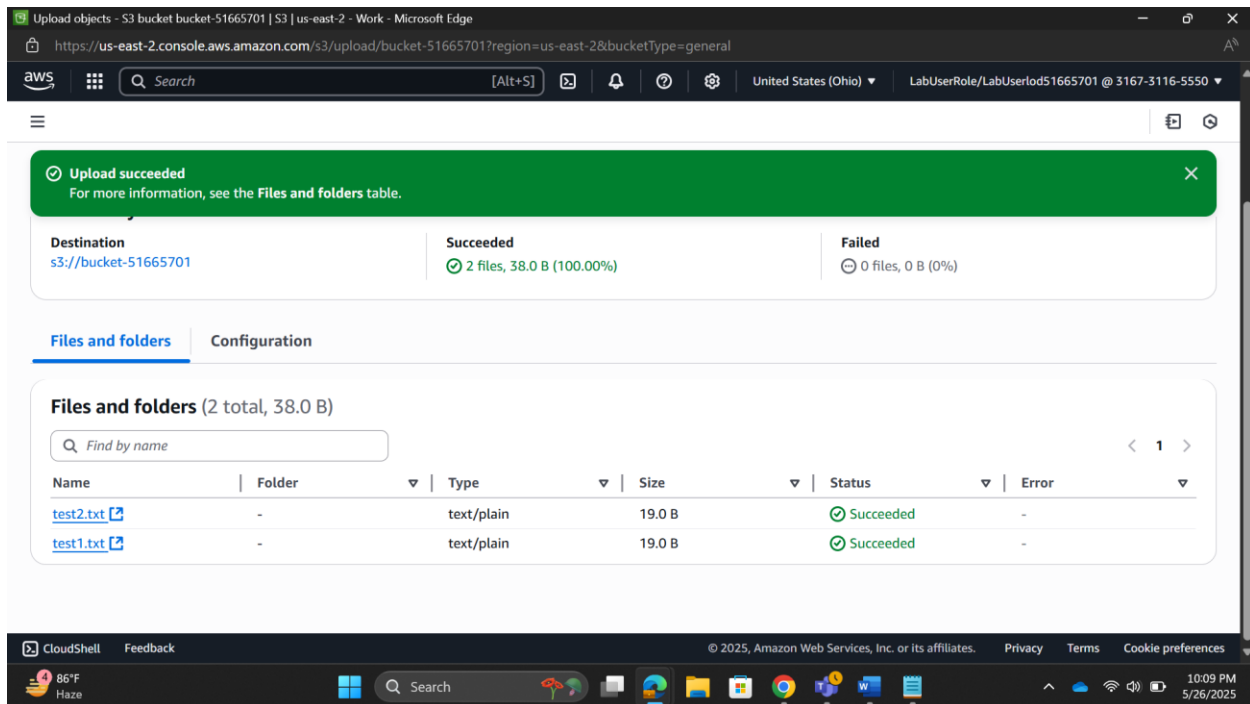
Scroll down click create bucket



Upload 2 txt files in bucket. Can use Notepad for creating these files

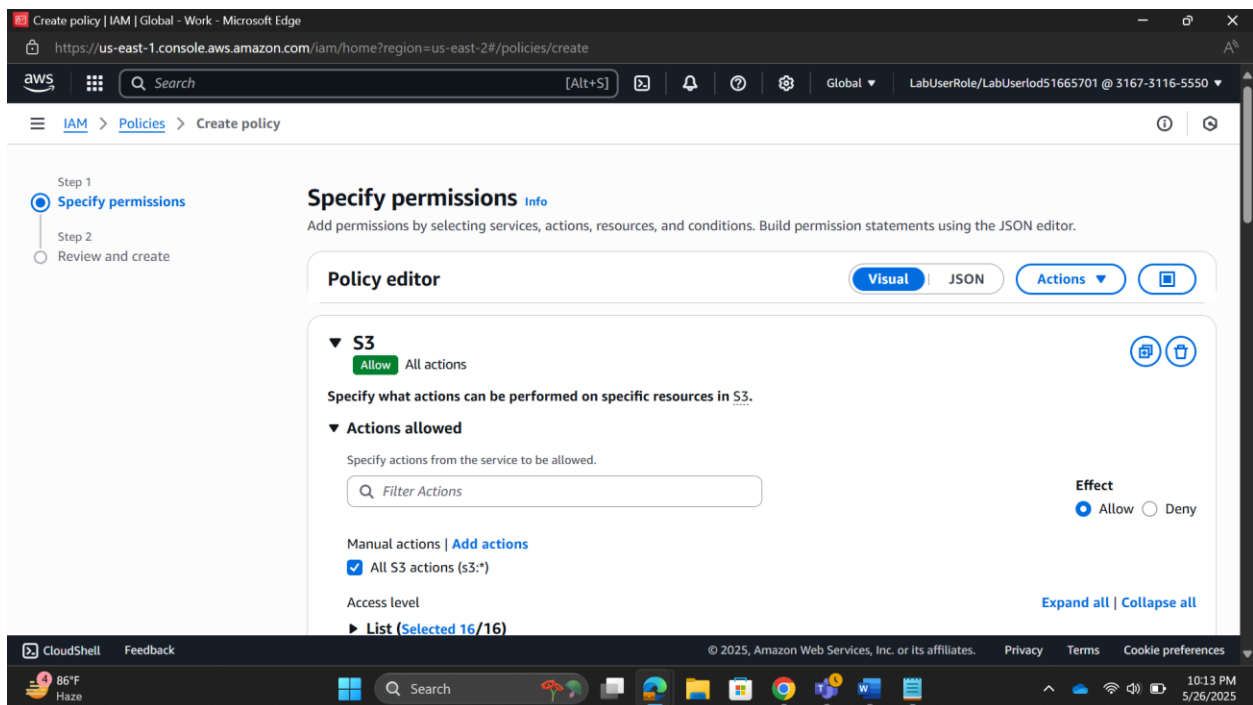
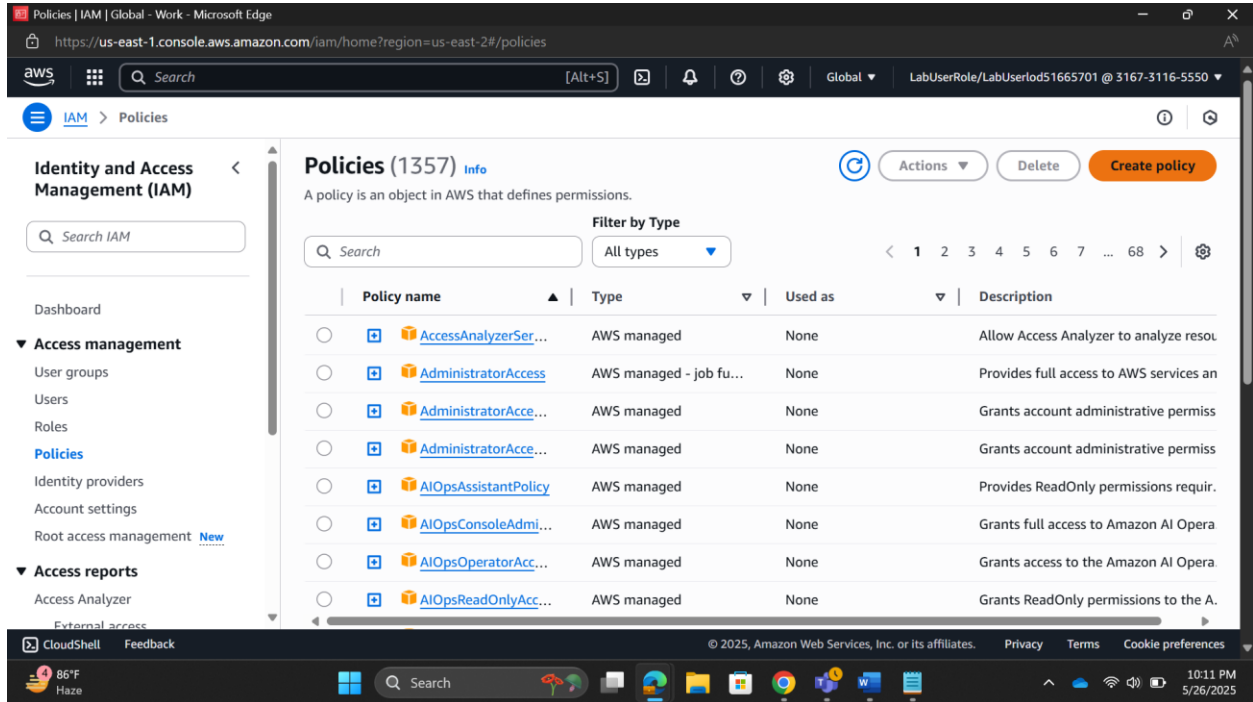
File 1: test1.txt = This is test file 1.

File 2: test2.txt = This is test file 2.

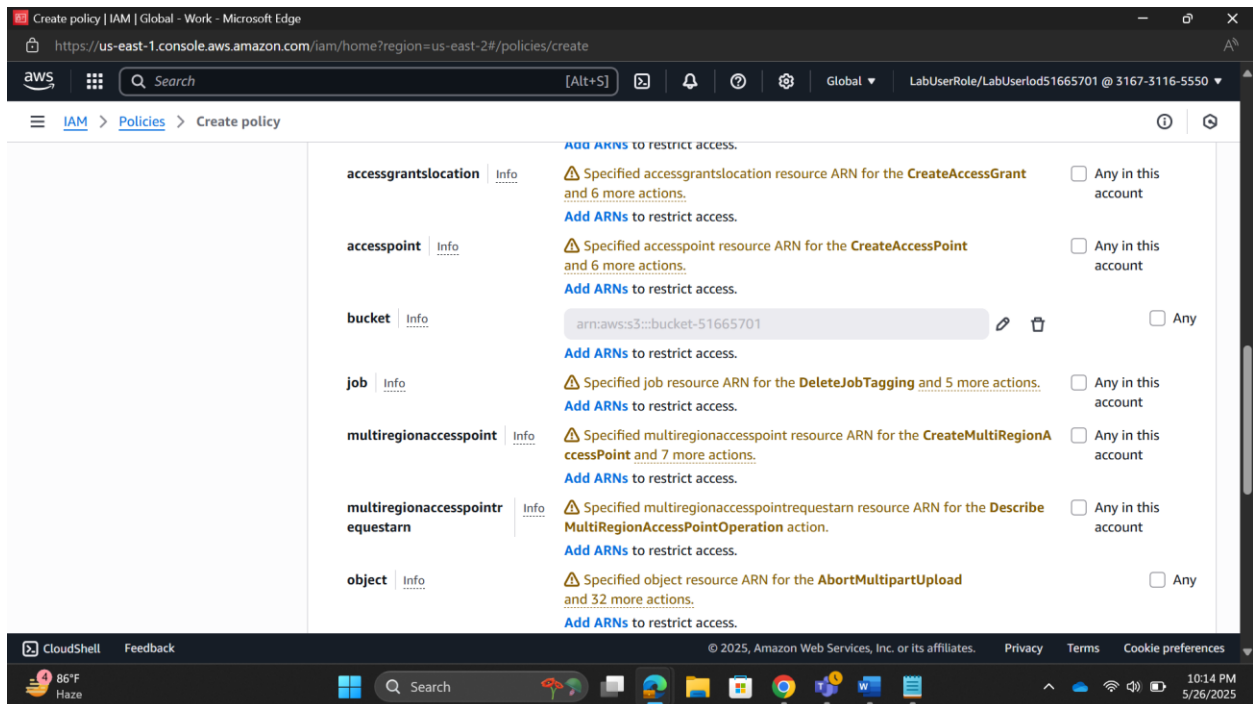
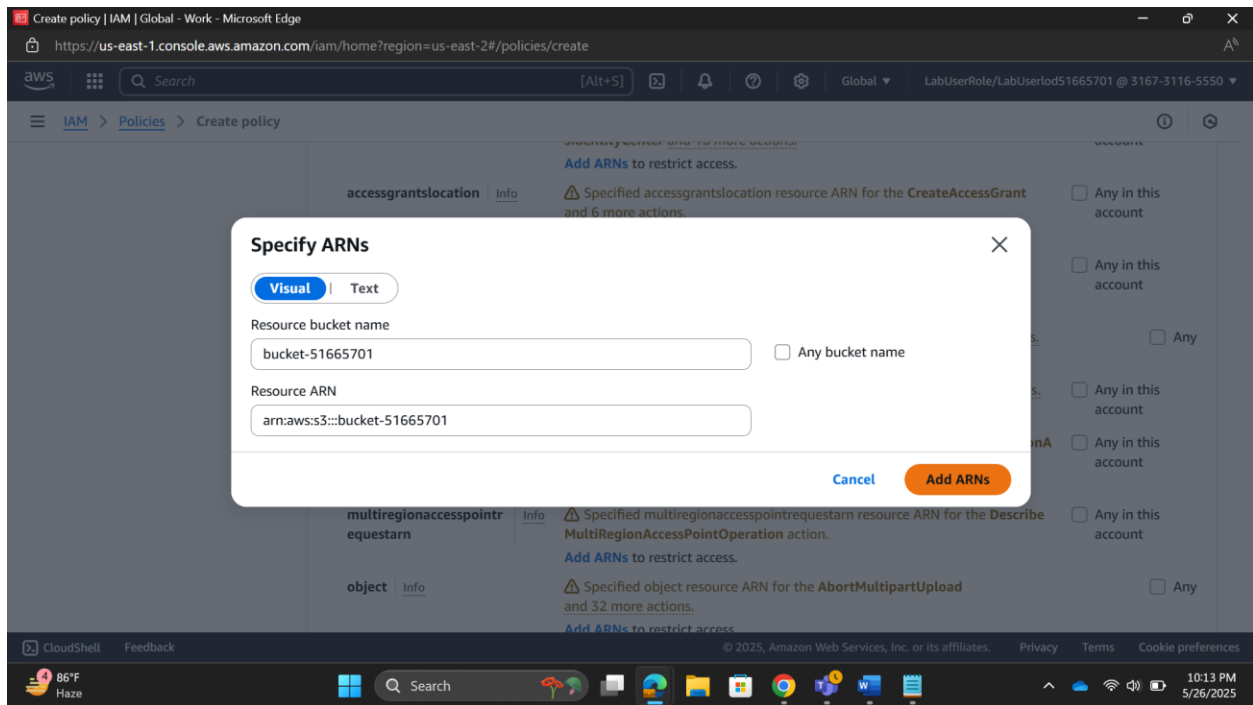


Task 2: Create a Policy

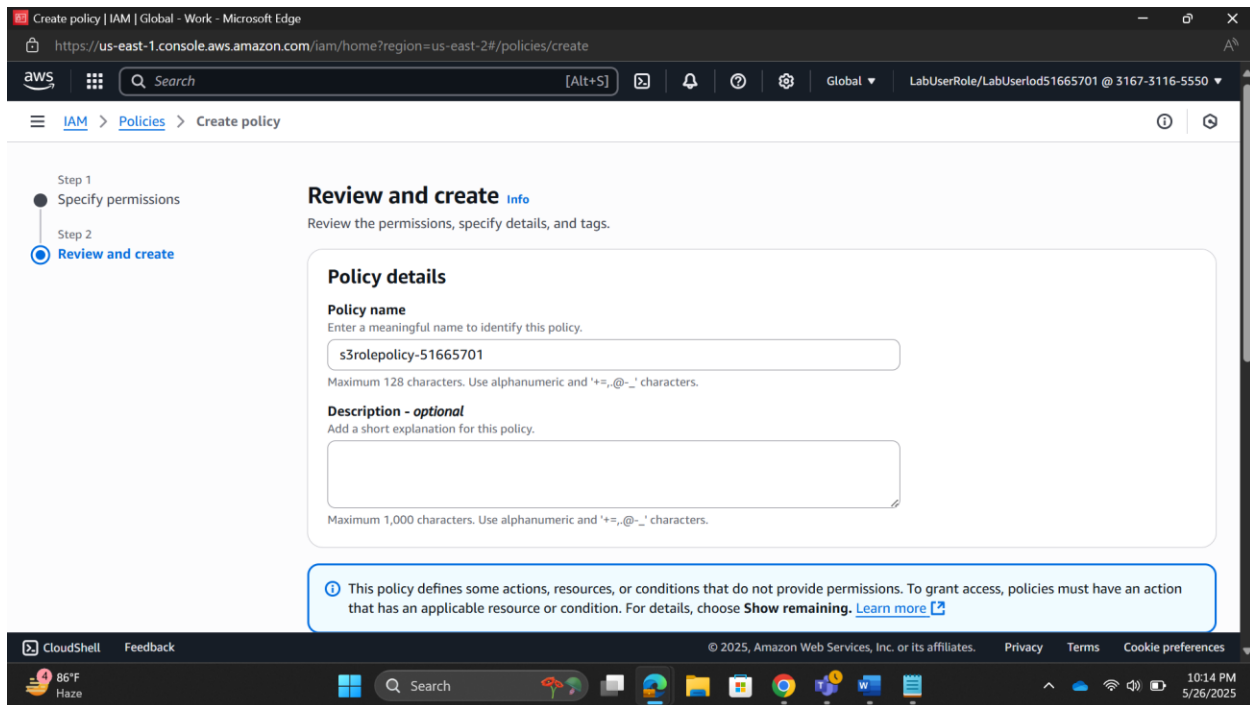
IAM -> access management -> policies -> create policy



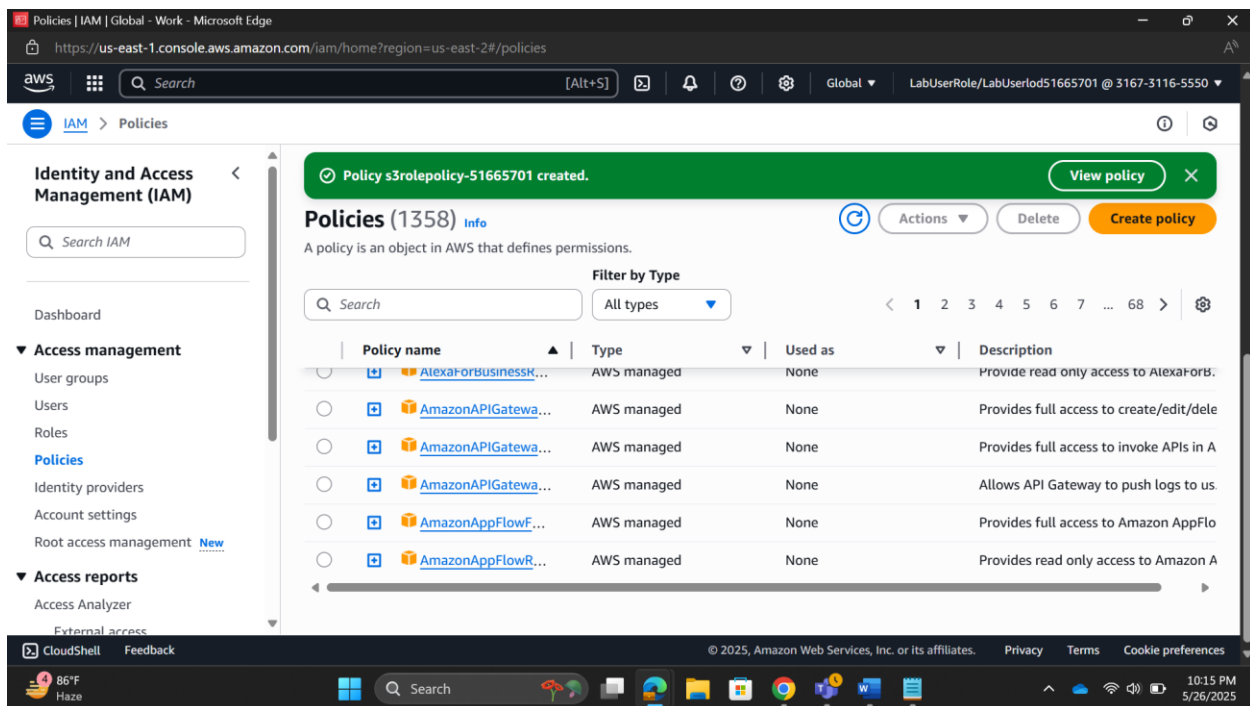
In Bucket add ARN



Click next



Click create policy



Task 3: Create a role to allow access to the S3 bucket

IAM -> Access management -> roles -> create roles

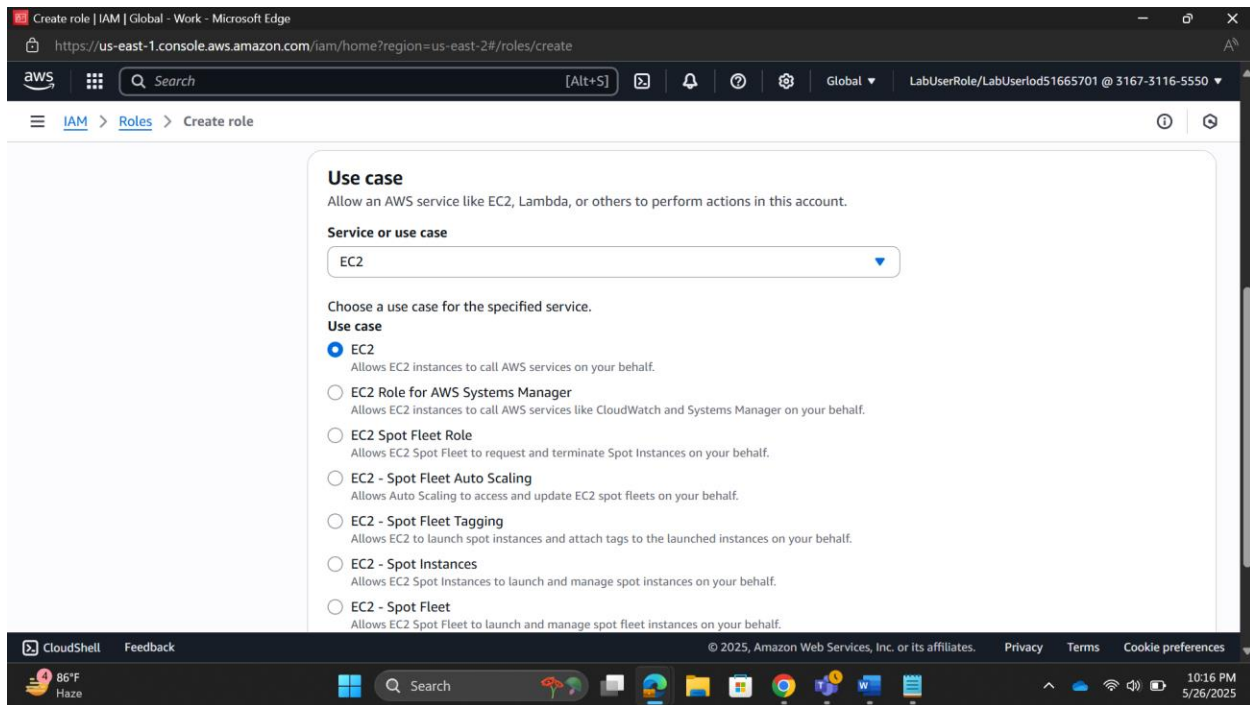
The top screenshot shows the AWS IAM console 'Roles (9)' page. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like Dashboard, Access management, Access reports, and External access. The main content area shows a list of roles with columns for Role name, Trusted entities, and Last activity. The roles listed are:

Role name	Trusted entities	Last activity
admin-function-execution-role	AWS Service: lambda	3 minutes ago
AWSServiceRoleForApplicationAutoScaling_DynamoDBTable	AWS Service: dynamodb.application	881 days
AWSServiceRoleForOrganizations	AWS Service: organizations (Service-Linked Role)	-
AWSServiceRoleForSSO	AWS Service: sso (Service-Linked Role)	-
AWSServiceRoleForSupport	AWS Service: support (Service-Linked Role)	751 days
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linked Role)	751 days
LabUserRole	Account: 927479171129	9 minutes ago
OrganizationAccountAccessRole	Account: 927479171129	13 minutes ago

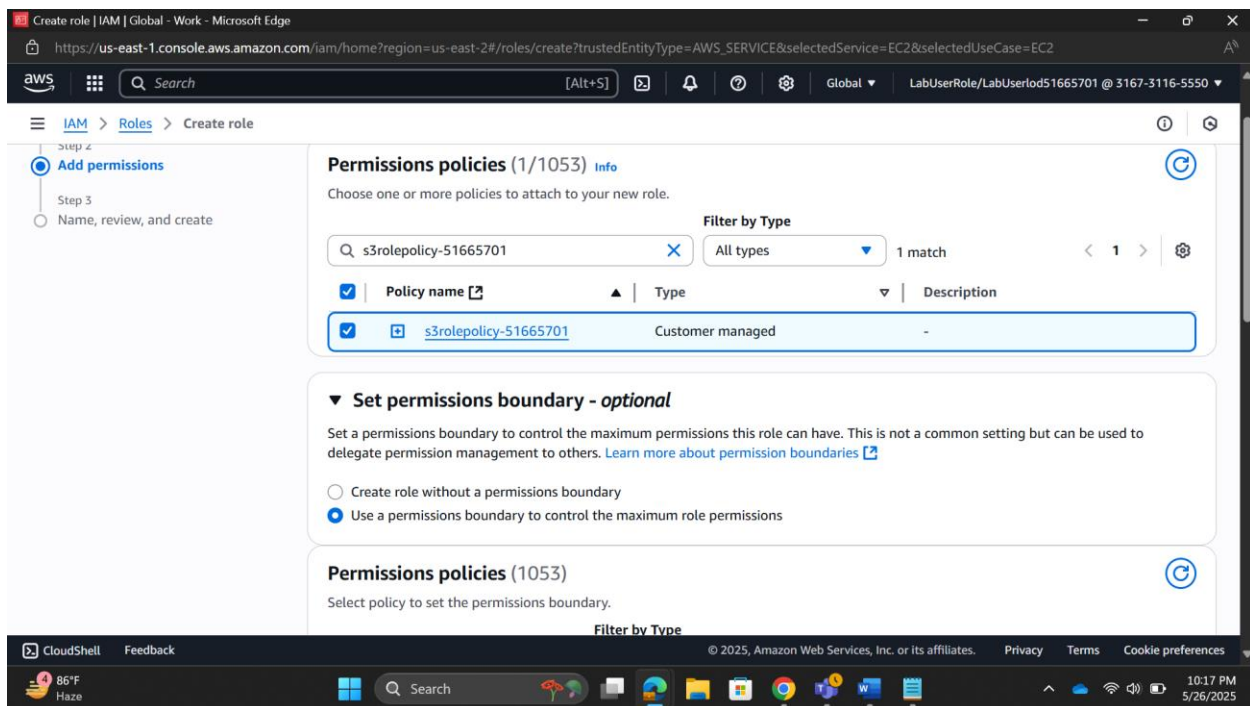
The bottom screenshot shows the 'Create role' wizard, Step 1: Select trusted entity. The 'Trusted entity type' section has five options:

- ☒ **AWS service**: Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**: Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**: Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**: Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**: Create a custom trust policy to enable others to perform actions in this account.

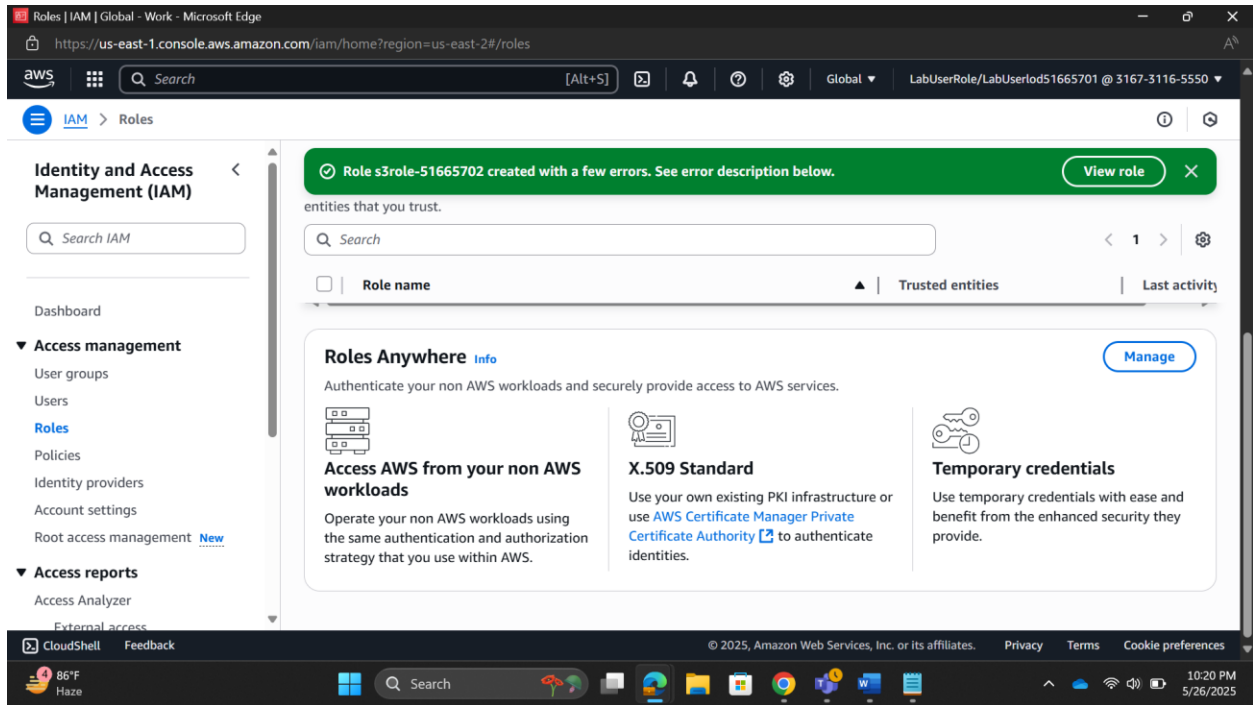
The 'Use case' section is currently empty.



Click next





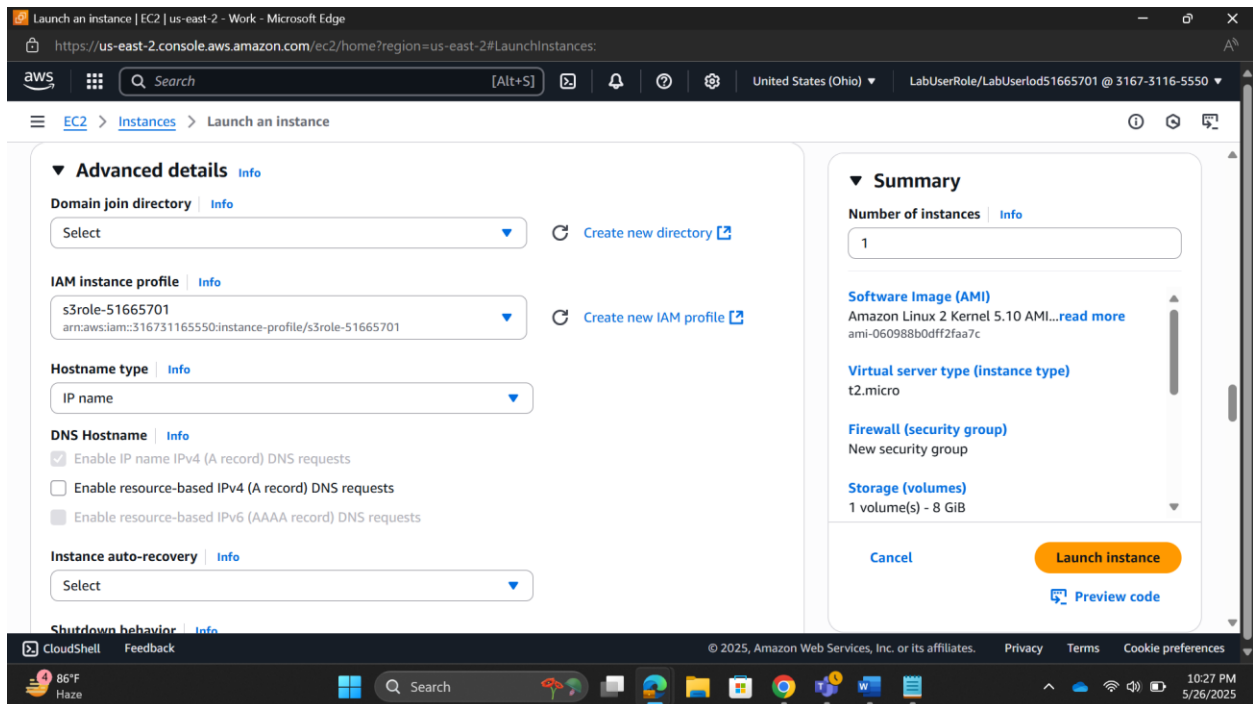
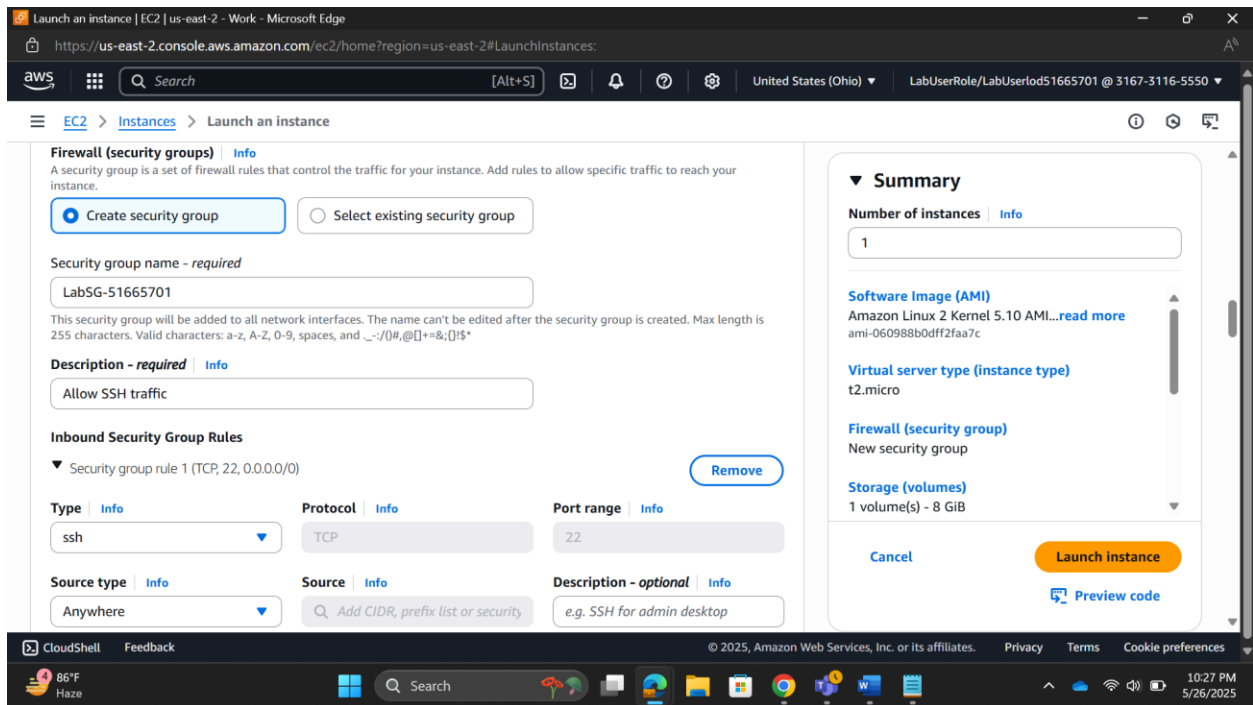


Task 4: Create an EC2 instance

Create EC2 instance with these details

- Begin configuring an Amazon EC2 instance by using the values in the following table. For any property that is not specified, use the default value.

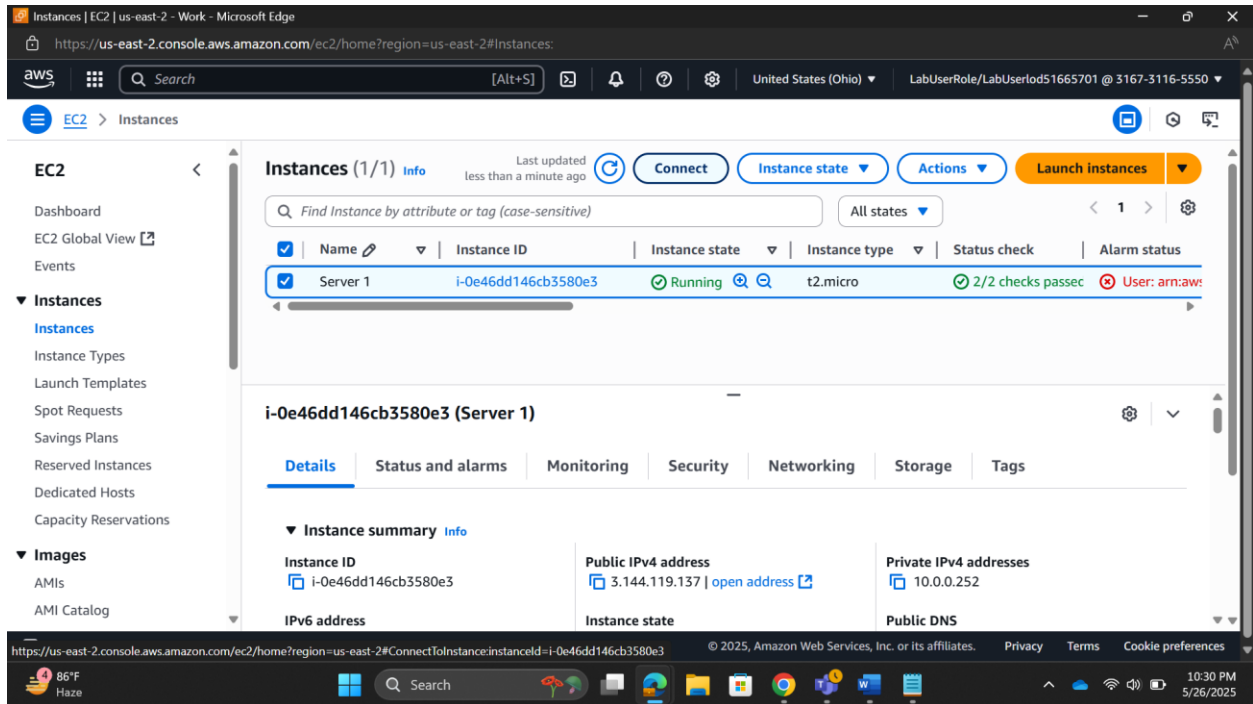
Property	Value
Name	Server 1
AMI	Amazon Linux 2 AMI (HVM), SSD Volume Type, 64-bit (x86)
Instance Type	t2.micro
Number of instances	1
Key pair	Proceed without a key pair
Network	Lab VPC A
Subnet	Public Subnet A1 us-east-2a
Metadata version	V1 and V2 (token optional)



Click launch instance

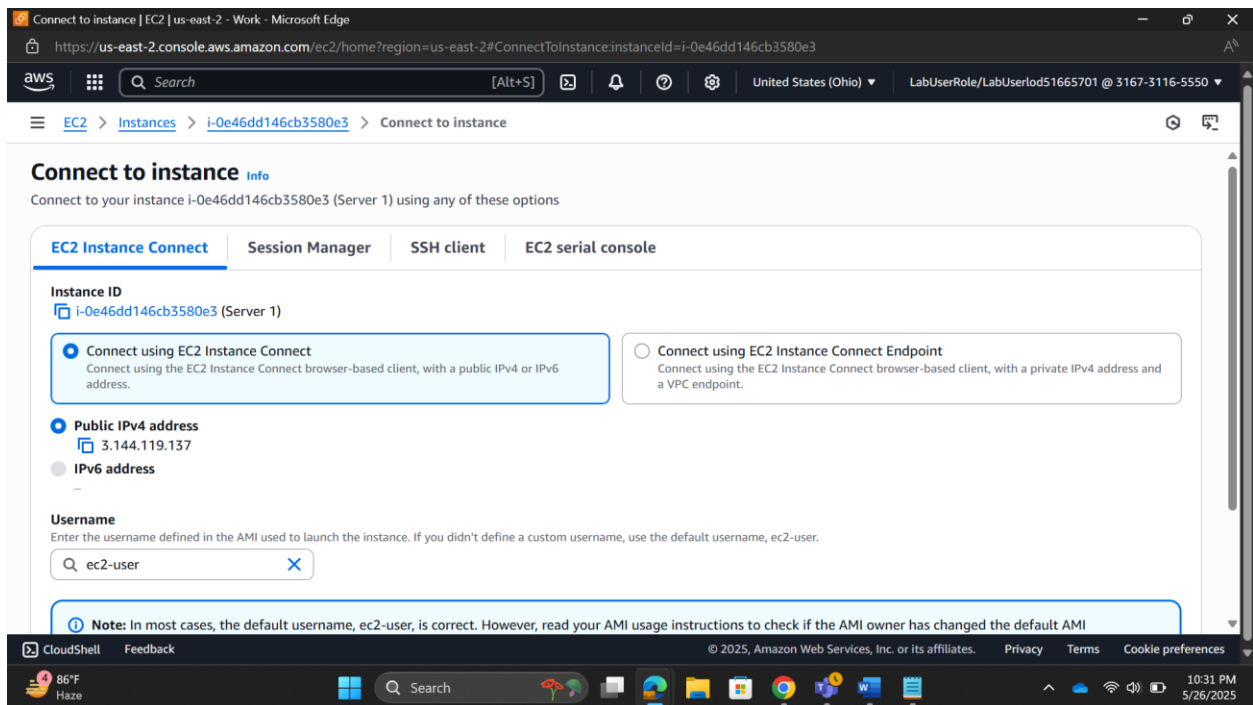
Wait till 2/2 checked

Task 5: Test access to the S3 bucket from the EC2 instance



The screenshot shows the AWS Management Console for the 'us-east-2' region. The 'Instances' page displays a table with one instance, 'Server 1', with ID 'i-0e46dd146cb3580e3'. The instance is in the 'Running' state. The 'Connect' button is highlighted. Below the table, the 'Details' tab for instance 'i-0e46dd146cb3580e3 (Server 1)' is selected, showing the 'Instance summary' with fields for Instance ID, Public IPv4 address (3.144.119.137), Private IPv4 addresses (10.0.0.252), IPv6 address, Instance state, and Public DNS.

Select instance -> click connect



The screenshot shows the 'Connect to instance' page in the AWS Management Console. The 'EC2 Instance Connect' tab is selected. The page displays the instance ID 'i-0e46dd146cb3580e3 (Server 1)'. Under the 'Connect using EC2 Instance Connect' section, the 'Public IPv4 address' is shown as '3.144.119.137'. The 'Username' field is set to 'ec2-user'. A note at the bottom states: 'Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI'.

Click connect

